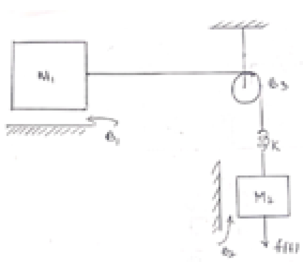
CONTROL SYSTEMS – UE24EE241B

Sample MCQs

ANALOGOUS SYSTEMS	
1.	**Analogous systems are those which have:** A. Same physical components B. Same mathematical model C. Same energy source D. Same output only
2.	**Which of the following is an example of analogous systems?** A. Electrical and thermal systems B. Mechanical and hydraulic systems C. Electrical and mechanical systems D. All of the above
3.	In force–voltage analogy, the **mechanical impedance** $$Z_m(s) = Ms + B + \frac{K}{s}$$ is electrically analogous to: A. $R + sL + \frac{1}{sC}$ B. $sL + R + \frac{1}{sC}$ C. $\frac{1}{R} + \frac{1}{sL} + sC$ D. $sC + R + \frac{1}{sL}$

4.	A translational mechanical system is modeled using **force–voltage analogy**. The governing equation is: $$F(t) = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx$$ The equivalent electrical circuit will contain: A. Capacitor, resistor, and inductor in **parallel** B. Capacitor, resistor, and inductor in **series** C. Only resistor and capacitor D. Only resistor and inductor
5.	The electrical analog of a mechanical system has an **inductor of value 2 H**, resistor of **4 Ω**, and capacitor of **0.5 F** in series. Using force–voltage analogy, the mechanical parameters are: A. $M = 2,\ B = 4,\ K = 2$ B. $M = 2,\ B = 0.25,\ K = 2$ C. $M = 0.5,\ B = 4,\ K = 2$ D. $M = 2,\ B = 4,\ K = 0.5$
6.	A mechanical impedance is given by: $$Z_m(s) = 5s + 10 + \frac{20}{s}$$ The equivalent electrical impedance (force–voltage analogy) is: A. $5s + 10 + \frac{20}{s}$ B. $10s + 5 + \frac{1}{20s}$ C. $\frac{1}{5s} + 10 + 20s$ D. $20s + 5 + \frac{10}{s}$

7.	A mechanical mass–spring–damper system is converted into its electrical analog **analogy.** Given: • $M = 2$ • $B = 4$ • $K = 8$ Which of the following are **correct**? A. $C = 2$ B. $R = 0.25$ C. $C = 0.5$ D. $R = 4$
8.	Identify the number of displacements for the given mechanical system  ○ 4 ○ 2 ○ 3 ○ 1
9.	A mechanical system modeled using force–current analogy has: $$\frac{X(s)}{F(s)} = \frac{1}{s(Cs^2 + \frac{s}{R} + \frac{1}{L})}$$ Which of the following are true? A. System contains a mass B. System contains a damper C. System contains a spring D. System is third order

Date: 27-01-26 **Dr. Anuradha M**
(Anchor)